Int. J. Human–Computer Studies 170 (2023) 102956

Contents lists available at ScienceDirect

International Journal of Human - Computer Studies

journal homepage: www.elsevier.com/locate/ijhcs

How compliance with web accessibility standards shapes the experiences of users with and without disabilities

Beat Vollenwyder [a,*], Serge Petralito [b], Glena H. Iten [c], Florian Brühlmann [a], Klaus Opwis [a], Elisa D. Mekler [d]

[a] Center for Cognitive Psychology and Methodology, Department of Psychology, University of Basel, Basel, Switzerland
[b] University of Applied Sciences and Arts Northwestern Switzerland, Olten, Switzerland
[c] Schwyz University of Teacher Education, Schwyz, Switzerland
[d] Digital Design Department, IT University of Copenhagen, Copenhagen, Denmark

ARTICLE INFO

Keywords:
Web accessibility
Usability
User experience
Users with visual impairments
Nondisabled users

ABSTRACT

Benefits for all user groups is one of the most prominent motivations to provide accessible information and services on the web. Designing digital technologies in a more inclusive manner for users with sensory, motor, or cognitive impairments enhances their overall quality. In practice, work on web accessibility often relies on complying with standards. But whether standards lead to improved usability and a satisfying user experience for all user groups is controversial. The present study aims at deepening our understanding of how compliance with web accessibility standards shapes the experiences of both users with and without disabilities. In a randomised controlled experiment, 66 participants with visual impairments and 65 participants without visual impairments solved tasks on an online shop built with either low (NA) or high (AA) conformance to web accessibility standards. The results show no statistically significant effects on outcomes related to usability and user experience. However, analysis of open-ended answers suggests that participants with visual impairments reported more positive experiences, and participants without visual impairments fewer negative experiences while using the online shop conformant to web accessibility standards. We therefore recommend adopting a more differentiated perspective on what can be achieved through compliance with web accessibility standards and emphasise that conformance-based approaches should be complemented with user-centred and participatory design methods. Further, since most participants reported being experienced users and an online shop is often a familiar context, more research in other settings is required.

1. Introduction

Digital technologies have become pervasive and are indispensable in many domains of everyday life, ranging from daily social, economic, and political participation. This development presents opportunities to include users with diverse abilities, aptitudes, and attitudes (Horton and Quesenbery, 2014). Web accessibility supports diversity and inclusion by preventing the emergence of barriers to using the web that can result from sensory (e.g., vision, hearing), motor (e.g., tremor, limited use of hands), and cognitive (e.g., learning disabilities, attention deficits) impairments. It aims to provide useable information and services to as many people as possible (Kalbag, 2017) and contributes to the web's role in enabling and promoting equal participation in society (Henry, 2006).

One of web professionals' main motivations for considering web accessibility is that it enhances the overall quality of a product (Vollenwyder et al., 2019; Yesilada et al., 2012). In daily work on web accessibility, such efforts are often closely tied to compliance with given standards (Holliday, 2020). Frequently used manual and automated evaluation methods are based on web accessibility standards (Abou-Zahra, 2008). While standards are undeniably a solid foundation for creating accessible information and services on the web (Henry, 2006), there have been contrary findings regarding their impact on issues related to usability and user experience. A differentiated understanding

* Corresponding author.
E-mail address: beat.vollenwyder@unibas.ch (B. Vollenwyder).

https://doi.org/10.1016/j.ijhcs.2022.102956
Received 26 March 2022; Received in revised form 26 August 2022; Accepted 17 October 2022
Available online 29 October 2022
1071-5819/© 2022 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

of how an emphasis on compliance may affect the users is therefore an essential step in the process of considering web accessibility.

2. Background

2.1. Web accessibility and standards

Providing an accessible and useable web to people with disabilities has been considered essential since the early development of this technology (World Wide Web Consortium, 1997), but a widely used and agreed-upon definition of web accessibility has not been established (Petrie et al., 2015). According to an industry survey (Yesilada et al., 2012), the most popular definition is provided by the Web Accessibility Initiative (WAI), a subgroup of the World Wide Web Consortium (W3C). It states, "Web accessibility means that people with disabilities can use the Web. More specifically, Web accessibility means that people with disabilities can perceive, understand, navigate and interact with the Web, and that they can contribute to the Web" (Yesilada et al., 2012, p. 2). Whereas this description has a specific focus on people with disabilities, another prominent definition introduced by the International Standard Organisation (ISO) broadens the perspective to "the usability of a product, service, environment or facility by people with the widest range of capabilities" (International Organization for Standardization, 2019, Chapter 3.1). These examples show that the concept of web accessibility overlaps to some extent with the related concepts of usability and user experience. Whereas usability focuses on the objective (effectiveness, efficiency) and subjective (satisfaction) outcome of an interaction in a specific context, user experience introduces a holistic view (anticipated and actual use) and a primary focus on emotions (Sauer et al., 2020). All of these aspects have been taken into account as the concept of web accessibility has evolved (e.g., Aizpurua et al., 2016; Petrie and Kheir, 2007; Shneiderman, 2000).

The development of the theoretical perspective on web accessibility with influences related to usability and user experience, is also reflected in the practical work in the field. Early work on web accessibility was mainly technically oriented and concentrated on providing basic access to web technologies. Power et al. (2018) refer to this phase as "First Wave Inclusion", where the main goal was to enable users to obtain and enter information in an interactive system. Providing basic access is an ongoing task and the foundation for the accessibility of each emerging technology. It is achieved by offering alternate input modes (e.g., switch access that replaces interaction via touchscreen for users with limited dexterity in their hands) and by translating information into alternate output modalities (e.g., access by screen reader for users with visual impairments). The development of such approaches combined with the rise of web use led to the publication of the first Web Content Accessibility Guidelines (WCAG 1.0), which consists of a checklist meant to support developers with regard to what they should and should not include in their implementations on the level of code (Power et al., 2018; Chisholm et al., 1999). These guidelines were later revised and rearranged with the publication of the second edition of the Web Content Accessibility Guidelines (WCAG 2.0). To be more future-proof, the revised criteria were formulated in a technology-agnostic manner and arranged around the four pillars that technology should be (1) perceivable, (2) operable, (3) understandable, and (4) robust, which are referred to as the POUR principles (Caldwell et al., 2008; Power et al., 2012). They provide a foundation for a shared understanding regarding the needs of users with disabilities and serve as a reference for specific design decisions and as a standard for evaluations (Harper and Yesilada, 2008; Horton et al., 2015). These functions make the guidelines vital for creating accessible information and services on the web (Henry, 2006). Today, WCAG 2.0 is the de facto standard for web accessibility and is referenced in the policies of several countries worldwide (Waddell, 2006).

2.2. Web accessibility and usability

With the establishment of WCAG 2.0, less technically oriented aspects of web accessibility have also received more attention (Power et al., 2018). By addressing typical usability-related criteria such as effectiveness, efficiency, and satisfaction in a specified context of use, the revision of the guidelines adapted and introduced criteria that centre on user tasks rather than on providing basic access (International Organization for Standardization, 2019, Chapter 3.13). This shift in perspective is referred to as "useable accessibility" (Web Accessibility Initiative, 2016) or "Second Wave Inclusion" (Power et al., 2018). One example for such a transition can be found in criterion 1.3.2, which demands a "meaningful sequence" in content presentation (Caldwell et al., 2008). For instance, if a document presents text in multiple columns, the implementation should convey that the content flows from the top of a column to the bottom and then to the top of the next column. A user with a screen reader can then perceive the document in a meaningful sequence. This requires a design decision and implies that the reading sequence must be centred around users and their goals in a specific situation, rather than on just having a technologically sound implementation (Power et al., 2018). Accessibility experts widely agree that such usability-related considerations must be part of web accessibility (Yesilada et al., 2014).

A broader perspective on web accessibility also expands the potential group of users affected by a given criterion. For example, standards for minimal contrast may be indispensable for users with visual impairments to interact with a product, but users without a visual impairment can also benefit from such standards in many temporary states (e.g., tired eyes after a long work day) or situations (e.g., using a device in strong sunlight). These considerations reflect the fundamental idea that web accessibility should extend the use of a product to as many people as possible (Kalbag, 2017). This favourable attitude regarding web accessibility is a prevalent opinion among web professionals, who often state that increasing the overall quality of a product is one of the main motivations for considering web accessibility (Vollenwyder et al., 2019; Yesilada et al., 2012). Further, it is in line with related concepts such as design for all (e.g., Bendixen and Benktzon, 2015), inclusive design (e.g., Clarkson et al.) and universal design(e.g., Iwarsson and Ståhl, 2009), which emphasise the idea of designing artefacts that address users with a wide range of capabilities (Sauer et al., 2020).

However, to what degree the compliance with given standards is sufficient to provide useable accessibility for all user groups is not without controversy in research and practice. User evaluations of 16 websites based on WCAG 2.0 showed that a large part of the problems encountered are either not solved by the given criteria or not covered at all (Power et al., 2012). Further, the often technical perspective of WCAG 2.0 is criticised as not centred on actual user needs (Cooper et al., 2012; Pereira and Archambault, 2018; Rømen and Svanæs, 2011) and as difficult to evaluate (Brajnik et al., 2012, 2011). This contrasts with a series of studies by Schmutz et al. (2016, 2017b,a), who invited participants with and without impaired eyesight to test a municipality's website in three different conformance levels. They measured indicators related to usability and user experience such as task-completion time, perceived usability, and the perceived visual aesthetic. The findings in all the studies showed a stable beneficial effect of high conformance with web accessibility standards. This effect was observed to the same extent for users with and without disabilities. However, all three studies used the same website and similar tasks, which may limit the degree to which these results can be generalised to other contexts. In sum, current findings leave an unclear picture of whether compliance with web accessibility standards alone is sufficient to positively influence usability-related measures.

2.3. Web accessibility and user experience

A focus on providing basic access and on usability-related factors lays a solid foundation for the use of digital technologies. Work on these topics has enabled users with and without disabilities to achieve their goals in many everyday situations. Yet as digital technologies have become more prevalent in all corners of our lives, non-task-oriented aspects have also gained importance. Not only do the pragmatic qualities of a product receive attention but also the hedonic qualities, which address human needs such as stimulation, relatedness, competence, and popularity (Hassenzahl et al., 2010). Such considerations in relation to web accessibility are part of "accessible user experience" (Web Accessibility Initiative, 2016) or "Third Wave Inclusion" (Power et al., 2018), with its user experience-oriented understanding of access to digital technologies. Since web accessibility wants to include as many people as possible, user needs become more diverse. The goal is therefore to design experiences that are not identical but comparable in value and quality for each user (Swan et al., 2017).

As for usability-related criteria, issues related to user experience are also affected by web accessibility considerations. In this context, web professionals and other stakeholders express more reservations. For example, web accessibility standards are feared as being a constraint on innovation, which leads to boring products that cannot keep up with current trends (Ellcessor, 2014). Another prominent overlap of web accessibility and user experience is in the aesthetic of digital products. Often named fears refer to the restrictions on contrast and the use of colour, which are perceived as compromising the visual design (Mbipom and Harper, 2011; Petrie et al., 2004). On the contrary, investigations regarding the relationship between web accessibility and the visual aesthetic of websites show that visual cleanness correlates with accessible content, whereas other dimensions, such as expressive design, seem to be neither a benefit nor a barrier to web accessibility (Mbipom and Harper, 2011). Another field of interest related to user experience in the context of web accessibility is how language simplification on the web may be perceived by users with and without cognitive disabilities (Schmutz et al., 2019; Vollenwyder et al., 2018; Karreman et al., 2007). Applying the concept of easy-to-read language, which attempts to be as simple as possible, results in improvements on some performance measures for all user groups but also decreased liking of the text and reduced intention to revisit a website for users with no identified cognitive disabilities. However, it is important to note that easy-to-read language was explicitly designed to meet the needs of people with cognitive impairments and was not intended to be a substitute for conventional language (Fajardo et al., 2013).

Studies exploring the impact of compliance with web accessibility standards on user experience are rarely available. Aizpurua et al. (2015) invited eleven blind participants to interact with a selection of websites that conformed with standards to various degrees. They concluded that compliance does not always correspond with users' actual perception of web accessibility and that factors such as prejudices, memories, expectations, and confidence in a website play an important role. Further, the authors also investigated how perceived web accessibility correlates with measurements of pragmatic and hedonic qualities (Aizpurua et al., 2016) and proposed applying such measures as a proxy to estimate conformance to web accessibility standards. Other studies have included measures related to user experience in experimental setups that compare conformant with non-conformant websites. Findings have shown a positive influence on the scales for affect, the aesthetic, and user experience (Schmutz et al., 2017b) as well as on mood (Pascual et al., 2014b) for users with visual impairments. Positive effects on mood have also been reported for users with hearing (Pascual et al., 2014a) and motor (Pascual et al., 2015) impairments. As for outcomes related to usability, study designs that included nondisabled participants (Schmutz et al., 2017b; Pascual et al., 2014b) have shown that this effect on experience is not only limited to users with disabilities but also benefits users without disabilities. All in all, the majority of current findings have ascertained a positive influence from compliance with web accessibility standards on measures related to user experience.

2.4. Aim of the study

As summarised above, the current research does not give a consistent answer as to whether the positive effects from compliance with web accessibility standards can be expected for outcomes related to usability and user experience. Despite this uncertainty, the most predominantly used evaluation methods for web accessibility rely on compliance with standards (Sauer et al., 2020). For example, expert-based manual testing (Abou-Zahra, 2008), barrier walkthroughs (Yesilada et al., 2009), and automated testing (Vigo et al., 2013) use given standards as their foundation. Further, it has become a common goal for organisations to obtain some form of label or certification from external evaluators to document their efforts in providing accessible information and services. Again, such certification processes are heavily dependent on the frame provided by web accessibility standards (Abou-Zahra, 2008). Although it is not a new demand to focus on the user perspective while working on web accessibility (e.g., Cooper et al., 2012; Henry, 2007; Karreman et al., 2007), the outlined overemphasis on standard conformance has become a common complaint by web professionals. In the worst case, web accessibility is experienced as a meaningless box-ticking exercise (Holliday, 2020; Oswal, 2019) that leads to poor implementations for all users.

The present study aims to contribute to a more differentiated understanding of how compliance with web accessibility standards shapes the experiences of users. Participants with and without visual impairments interacted in a randomised controlled experiment with either a conformant or a non-conformant version of a website. The materials used took the requirements of WCAG 2.0 into account. To complement previous studies, the materials were situated in the context of an online shop. The study design made it possible to explore the impact of compliance with web accessibility standards on outcomes related to usability and user experience.

3. Method

3.1. Participants and design

3.1.1. Sampling procedure

A total of 167 participants completed the online study. We targeted a sample size comparable to previous studies that reported significant group differences for compliant and non-compliant websites (see Schmutz et al., 2017b,a). Invitations were sent via three ways. (1) The study was advertised over various newsletters and mailing lists across Switzerland specifically targeting audiences with visual impairments. Further, (2) the affiliated institute's participant pool and (3) the recruiting service TestingTime[1] were used. As an incentive for the first two channels, participants could take part in a lottery for three gift vouchers of 30 Swiss Francs (about 27 Euros), while participants via TestingTime received a payment of 7 Swiss Francs (about 6.50 Euros).

3.1.2. Data cleaning

In total, 36 participants were subsequently removed from the sample. The applied criteria followed recommendations for data quality in online surveys (Brühlmann et al., 2020; Curran, 2016). Twenty-six participants were removed for using a smartphone to complete the study, even though the instructions stated to use a desktop or laptop computer. One participant declared that they did not answer the study seriously and that their data should not be used for the main analysis. Three participants were removed after we screened open-ended answers for obviously careless responses, such as the use of random words or copying a single answer. One participant was removed because of a response time that exceeded 90 min. Four participants were removed because their task completion time was below or above three standard deviations from the mean. Finally, one participant was removed because of a LongString Index that indicated that more than half of their questionnaire answers in a row were identical.

[1] https://www.testingtime.com/en

Table 1
Matching variables divided into the experimental conditions user group (impaired eyesight, unimpaired eyesight) and conformance with web accessibility standards (NA = low conformance, AA = high conformance).

	Impaired eyesight		Unimpaired eyesight		Total	
	Level NA	Level AA	Level NA	Level AA	Level NA	Level AA
N	29	37	35	30	64	67
Age (*SD*)	36.3 (15.2)	39.3 (14.4)	38.4 (11.1)	40.5 (11.5)	37.5 (13.1)	39.9 (13.1)
Computer experience (*SD*)	6.0 (0.9)	6.1 (0.9)	6.0 (0.9)	6.1 (1.0)	6.0 (0.9)	6.1 (0.9)
Web experience (*SD*)	6.2 (0.9)	6.3 (0.9)	6.0 (0.9)	6.1 (1.0)	6.1 (0.9)	6.2 (0.9)

3.1.3. Sample description

The study employed a 2 × 2 between-subjects design with the user group (impaired eyesight, unimpaired eyesight) and conformance to web accessibility standards (NA = low conformance, AA = high conformance) as independent variables.

A total of 131 participants (age $M = 38.7$, $SD = 13.1$, range 18–77; 63 women, 65 men, 3 non-binary or not specified) was included in the main analysis. An overview is presented in Table 1. Two groups of participants, one with visual impairments ($N = 66$) and without (65), formed the sample. Participants with visual impairments were required to have a maximum visual acuity of 0.3 (World Health Organization WHO, 2019). In this group, people with reduced eyesight ($N = 15$, visual acuity under 0.3), people with severe visual impairments (9, visual acuity under 0.05), people considered to be blind (1, visual acuity under 0.02) and people considered to be fully blind (15) took part in the study. Information on participants' eyesight was self-reported and no vision tests were conducted. As digital assistive technologies, participants used screen magnifiers (22, 33.3% of the participants with visual impairments), screen readers (17, 25.8%), screen readers combined with a screen magnifier (7, 10.6%), and screen readers combined with a braille board (5, 7.6%). On average, the study took 22 min ($SD = 13.5$, range 4.5–75) to complete. High self-rated experience with computers in general ($M = 6.1$, $SD = 0.9$) and with using websites in particular ($M = 6.2$, $SD = 0.9$) was reported (both 7-point Likert scale, 1 = very low, 7 = very high). About two-thirds of the participants ($N = 78$, 59.5%) were recruited via newsletters, mailing lists, and the affiliated institute's participant pool, whereas the other third (53, 40.5%) participated via the recruiting service TestingTime.

Age, gender, self-rated experience with computers and with using websites were tested statistically for differences between the experimental groups. No significant differences were found ($p > .05$ for all variables). Additional analyses were conducted for participants who dropped out of the study midway and who were not included in the main sample (N = 65). Only participants that interacted for at least 10 s with the study before dropping out were analysed ($N = 61$). Participant dropout was not more frequent in a particular condition or user group ($p > .05$ for all experimental conditions).

3.2. Materials

3.2.1. Websites and tasks

An online shop for organically produced vegetables was used as stimuli for the present study. Three reasons led to this choice: (1) online shops are reported among the most problematic services regarding web accessibility (Heim et al., 2020); (2) they are a common type of website and permit a selection of tasks well suited for the study demands; and (3) they complement previous studies that focused on websites for restaurants (Aizpurua et al., 2015, 2016), tourist information (Pascual et al., 2014b), mobile-phone companies (Petrie and Kheir, 2007), leisure centres (Schmutz et al., 2019), and municipal services (Schmutz et al., 2016, 2017a,b). The online shop in the present study offered a wide range of organically produced vegetables that were organised into various categories and described with details about their origin and season. Further, the website provided information about the terms and conditions and some background information about the farm that (virtually) produced the vegetables. It was assumed that the overall theme of sustainable and healthy nutrition is of importance to many people, which led us to choose this type of online shop. We prepared two versions of the online shop, one with low conformance (NA) to web accessibility standards and one with high conformance (AA). A version with basic conformance (A) was not considered because most legal obligations require level AA, and previous findings have not shown significant differences between level NA and A, nor between level A and AA (Schmutz et al., 2016, 2017b). As can be seen on the screenshots presented in Fig. 1, the conformant condition featured higher contrasts between text and background, clearer indication of the navigation state, and a more visible keyboard focus. Further, there were various differences that are not perceivable on a screenshot, such as coherent heading structures, alternative texts for images, and clear tab orders. We only altered aspects required by WCAG 2.0 (e.g., contrast and link descriptions), while other characteristics of the online shop (e.g., text content or pictures) remained identical in both conditions. Manipulations applied by Schmutz et al. (2016, see section"Website Manipulation"), were used as inspiration for the present study but adapted and extended for the online shop context. In total, 21 criteria were manipulated. A detailed description is provided in the Appendix. After creating the websites, we chose three tasks for the participants that included typical activities on an online shop. The tasks are described in Table 2.

To ensure realistic manipulations, both versions of the online shop were tested and discussed in a workshop with experts from the foundation Access for all,[2] a Switzerland-based non-profit organisation offering training, reviews, and certifications for web accessibility. Additionally, an automated evaluation using the Web Accessibility Evaluation Tool (WAVE[3]) was run on the home, product, product detail, information, and checkout pages. This test showed a total number of 48 errors for the low-conformance condition and 2 errors for the high-conformance condition. The complete study was then pretested by three blind participants. Besides the manipulations, the study setting itself, including the online survey tool, was also tested and deemed accessible.

3.2.2. Quantitative measures

Performance was measured by evaluating the task completion rate (%) according to the criteria defined in Table 2 and by tracking the time spent on the online shop as the task completion time (seconds). Subjective measures were assessed after each task and after the completion of all the tasks. After each task, participants were asked to provide an estimate of whether they had or had not succeeded (perceived task success; nominal scale with yes, no, do not know). Further, satisfaction after completing the task was assessed with the After Scenario Questionnaire (ASQ; Lewis, 1995). After the completion of all the tasks, additional subjective measures were assessed. To allow for comparisons, the selection of questionnaires was kept similar to previous studies in the field (see Schmutz et al., 2016, 2017a,b). The German versions of four measures were used. (1) Perceived usability was evaluated with the Website Analysis and MeasureMent Inventory (WAMMI; Kirakowski et al., 1998), whereas the standard 5-point scale was replaced with a 7-point scale for a consistent response format;

[2] https://www.access-for-all.ch/en
[3] https://wave.webaim.org

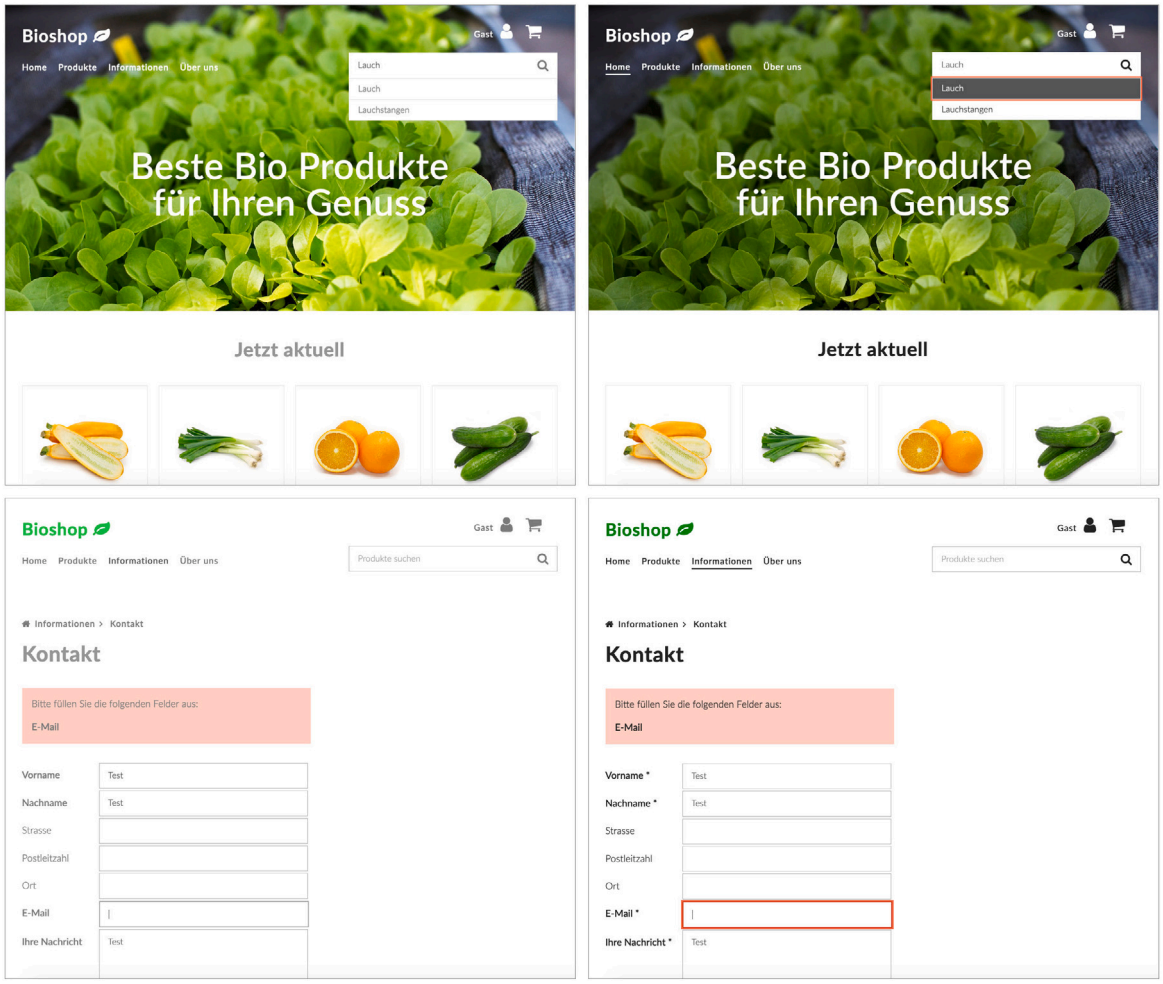


Fig. 1. Screenshots of the online shop used in the study. Top left: Home page with low conformance (NA) while searching for leeks. Top right: Home page with high conformance (AA). Bottom left: Contact form with low conformance while sending a request. Bottom right: Contact form with high conformance.

Table 2
Task descriptions.

Task	Description
1	Placement of an order on the online shop. Four items had to be found and added to the shopping cart, and the payment process had to be started. The task was successful if the correct items in the demanded amount had been added to the shopping cart when the checkout process was started.
2	Retrieval of information. Information regarding order closing times had to be found on the website. The task was successful if the correct time was reported.
3	Communication via contact form. A question had to be asked through the contact form of the shop. The task was successful if a message was sent from the contact form; the exact content of the message was not rated.

(2) the perceived aesthetic was measured with the short version of the Visual Aesthetics of Websites Inventory (VisAWI; Moshagen and Thielsch, 2013); (3) pragmatic and hedonic quality were rated with the short version of AttrakDiff2 (Hassenzahl and Monk, 2010); and (4) the likelihood to recommend the online shop was estimated by using the Net Promoter Score (NPS; Reichheld, 2003).

3.2.3. Qualitative measures

After the completion of each task and after the questionnaire section with subjective measures, participants were given the option to leave an open-ended comment ("If you would like to comment on this task (difficulties, uncertainties, etc.), please leave us a message".). Additionally, an open-ended question regarding overall difficulties with the website was included after the completion of all the tasks ("Have you encountered any obstacles while using the website? If so, please describe".). In total, we received 330 comments, sometimes describing multiple experiences. Responses were analysed with a qualitative content analysis (Flick et al., 2004). Following the definitions by Sauer et al. (2020), all reported experiences were assessed regarding their relation to web accessibility, usability, or user experience and their connotation with either positive or negative affect (see Table 3 for examples). Experiences that were not assignable to one of the concepts were analysed in a "not assignable" category, whereas experiences that were solely related to the study materials itself (e.g., comments regarding the survey tool, the pricing of the articles, etc.) were not included in the analysis. Sentences formed the smallest coding units and could be coded into multiple categories. The first author coded all open-ended answers. To ensure inter-rater reliability, a second independent

Table 3
Coding examples for open-ended answers.

Code	Affect	Example
Accessibility	Positive	"Could zoom in on the page well". (P125, NA condition, impaired eyesight)
	Negative	"The buttons to add a product to the shopping cart are missing labels". (P2, NA condition, impaired eyesight)
Usability	Positive	"One can find everything very quickly, read it clearly, and quickly order it". (P229, AA condition, impaired eyesight)
	Negative	"At first I couldn't find the vegetables that weren't on the start page. It took me a while to realise that there was a search field at the top". (P212, NA condition, impaired eyesight)
User experience	Positive	"I felt comfortable on the site and was able to read everything without a problem". (P204, AA condition, impaired eyesight)
	Negative	"The layout is professional. – Yes, but also a bit boring". (P17, AA condition, unimpaired eyesight)
Not assignable	Positive	"It was ok". (P224, NA condition, impaired eyesight)
	Negative	"I did not complete the order". (P111, NA condition, impaired eyesight)

coder coded a subset of 110 (33%) randomly selected comments. A substantial inter-rater agreement was achieved (Cohen's $\kappa = .67$), in line with Landis and Koch's recommendations (1977).

3.2.4. Manipulation check

In order to validate the experimental manipulation, an item asking for an accessibility rating ("Please rate the online shop according to its accessibility".; 1 = less accessible than other online shops, 7 = more accessible than other online shops) was included after the completion of all tasks. Participants unfamiliar with the term accessibility had the option not to answer. Since the website was specifically created for the present study, this item was used to estimate the external validation of the prepared materials, as well as to check for realistic manipulations. Reasonable experimental conditions would imply that participants rate the low-conformance (NA) version and the high-conformance (AA) version in a medium range in comparison to the quality of other typical online shops they have used in the past and therefore indicate authentic materials. Further, ratings between the two versions should not differ significantly, since this would indicate an unrealistic downgrade with barriers that would only appear if somebody intentionally breaks web accessibility. A comparable item was applied in a previous study (Schmutz et al., 2016). Further, open-ended answers were screened for accounts regarding the realism of the website and for unintended effects, such as technical problems with the website or the questionnaire.

3.3. Procedure

3.3.1. Study procedure

After informed consent was obtained, participants were asked to provide demographic information, to specify their eyesight, and, if applicable, to declare their setup of assistive technologies. Further, a self-rating of experience with computers ("How do you rate your experience with using computers?") and with the web ("How do you rate your experience with using the web?") was collected. Participants were then randomly assigned to the low-conformance (NA) condition or the high-conformance (AA) condition and asked to solve three typical tasks on an online shop as described in Table 2. The presentation of the tasks was counterbalanced to avoid order effects. For each task, participants were forwarded to the prototype of an online shop and returned to the questionnaire after solving or cancelling the task by choice. No login was required to access the online shop. The measurement of the task completion time started with the forwarding and ended with the return. No time limit was given. Task completion rate and completion time were automatically tracked during this procedure. A short questionnaire followed each task, and a longer questionnaire completed the study (see Section 3.2.2).

3.3.2. Missing data

For the ASQ, WAMMI, and VisAWI, participants were given the option to respond with "don't know" for a single item. This led to a total of 3.2% of missing data for these scales. To maintain as much information as possible, it was decided to impute the missing data. A visualisation of the data showed no noticeable patterns, and we assumed the data were missing at random. Expectation maximisation with age, gender, disability, experience with computers, and experience with websites as related variables was used to impute the missing data. The missing data were analysed using the naniar package (v0.6.1; Tierney et al., 2021), and the Amelia package (v1.8.0; Honaker et al., 2021) was used for data imputation.

4. Results

4.1. Manipulation check

Participants with visual impairments (number of ratings = 54, 81.8% of the sample) rated the low-conformance version ($M = 5.58$, $SD = 1.47$, 95% CI [4.96, 6.20]) as well as the high-conformance version ($M = 5.37$, $SD = 1.43$, 95% CI [4.83, 5.90]) in a medium range in comparison to the quality of other online shops they had used in the past (1 = less accessible than other online shops, 7 = more accessible than other online shops). Ratings between the versions did not differ significantly ($t(49) = 0.54$, $p = .588$). Similar results were reported by participants without visual impairments (number of ratings = 38, 58.5% of the sample) for the low-conformance version ($M = 5.24$, $SD = 1.37$, 95% CI [4.61, 5.86]) and the high-conformance version ($M = 4.94$, $SD = 1.14$, 95% CI [4.35, 5.53]), but no significant difference between the versions was found ($t(36) = 0.73$, $p = .472$). These results indicate that the used materials were perceived as comparable to an existing online shop and that the manipulations were not so extreme as to implement an unrealistic downgrade.

4.2. Quantitative measures

No significant differences were found for all the measures as a function of conformance to web accessibility standards. The user group and the interaction between conformance and the user group also did not exhibit any significance. The results of all the performance and subjective measures are presented in Table 4. As the use of various types of assistive technologies strongly influences how users navigate the web, we additionally analysed participants mainly using screen readers ($N = 29$, including combinations with magnifiers and braille boards) and participants mainly using screen magnifiers (22). Again, none of the measures did differ significantly for conformance with

Table 4
Results of performance and subjective measures.

Task completion rate (in %)						
	Level NA			Level AA		
	Mean	*SD*	Median	Mean	*SD*	Median
Impaired eyesight	66.7	33.3	66.7	73.9	34.4	100
Unimpaired eyesight	77.1	26.5	66.7	65.6	30.9	66.7
Total	72.4	30	66.7	70.1	32.9	66.7
Task completion time (in seconds)[1]						
	Level NA			Level AA		
	Mean	*SD*	Median	Mean	*SD*	Median
Impaired eyesight	81	57.8	63	103	53.5	97
Unimpaired eyesight	82	34.2	75.5	82.3	46.4	69.5
Total	81.6	45.3	69	93.5	51	85
Perceived task success (in %)						
	Level NA			Level AA		
	Mean	*SD*	Median	Mean	*SD*	Median
Impaired eyesight	93.1	20.7	100	95.5	14	100
Unimpaired eyesight	98.1	7.9	100	96.7	13.4	100
Total	95.8	15.1	100	96	13.6	100
Task satisfaction (ASQ, 1-7)						
	Level NA			Level AA		
	Mean	*SD*	Median	Mean	*SD*	Median
Impaired eyesight	4.9	2.3	6	5.2	2.2	6
Unimpaired eyesight	5	2	5.7	5.3	1.7	5.3
Total	4.9	2.1	6	5.2	2	6
Perceived usability (WAMMI, 1-7)						
	Level NA			Level AA		
	Mean	*SD*	Median	Mean	*SD*	Median
Impaired eyesight	5.5	1.4	5.8	5.4	1.4	6
Unimpaired eyesight	5.6	1.4	5.8	5.2	1.2	5.3
Total	5.5	1.4	5.8	5.3	1.3	5.9
Visual aesthetic (VisAWI, 1-7)[2]						
	Level NA			Level AA		
	Mean	*SD*	Median	Mean	*SD*	Median
Impaired eyesight	5.7	1.4	6	5.8	1.4	6
Unimpaired eyesight	5.6	1.6	6	4.8	1.3	4.9
Total	5.6	1.5	6	5.3	1.4	5.8
Pragmatic quality (AttrakDiff, 1-7)						
	Level NA			Level AA		
	Mean	*SD*	Median	Mean	*SD*	Median
Impaired eyesight	5.8	1.3	6	5.9	1.1	6.2
Unimpaired eyesight	6	1.2	6.2	5.6	1.2	5.8
Total	5.9	1.2	6.1	5.8	1.2	6
Hedonic quality (AttrakDiff, 1-7)						
	Level NA			Level AA		
	Mean	*SD*	Median	Mean	*SD*	Median
Impaired eyesight	5.1	1.4	5.2	5.1	1.3	5
Unimpaired eyesight	5	1.3	5	4.4	1	4.5
Total	5	1.3	5.1	4.8	1.2	4.8
Likelihood to recommend (NPS, total score of -100% to +100%)						
	Level NA			Level AA		
Impaired eyesight	10.3			21.6		
Unimpaired eyesight	-14.3			-30		
Total	-3.1			-1.5		

[1] Only successful attempts were used to compare task completion times.
[2] Fully blind participants were not asked for their ratings regarding the visual aesthetic.

web accessibility standards. The results focusing on the used assistive technologies are presented in Table 5.

Task completion rate. No significant differences in task completion as a function of conformance to web accessibility standards were found ($F(1, 127) = 0.17$, $p = .683$, partial $\eta^2 = .001$, 90% CI[0.00, 0.03]). Nor were significant differences found for the user group ($F(1, 127) = 0.03$, $p = .874$, partial $\eta^2 < .001$, 90% CI[0.00, 0.01]) or for the interaction between conformance and the user group ($F(1, 127) = 2.90$, $p = .091$, partial $\eta^2 = .022$, 90% CI[0.00, 0.08]).

Task completion time. Only successful attempts were used to compare task completion times. Results did not differ significantly for conformance ($F(1, 116) = 1.84$, $p = .178$, partial $\eta^2 = .016$, 90% CI[0.00, 0.07]), the user group ($F(1, 116) = 1.31$, $p = .255$, partial $\eta^2 = .011$, 90% CI[0.00, 0.06]), or the interaction between conformance and the

Table 5
Results of performance and subjective measures of participants mainly using screen readers ($N_{\text{Level NA}} = 12$, $N_{\text{Level AA}} = 17$) and participants mainly using screen magnifiers ($N_{\text{Level NA}} = 11$, $N_{\text{Level AA}} = 11$).

Task completion rate (in %)						
	Level NA			Level AA		
	Mean	SD	Median	Mean	SD	Median
Screen Reader	75	32.2	83.3	68.6	39.9	100
Screen Magnifier	60.6	36	66.7	72.7	36	100
Total	68.1	34.1	66.7	70.2	37.8	100
Task completion time (in seconds)						
	Level NA			Level AA		
	Mean	SD	Median	Mean	SD	Median
Screen Reader	73.8	69.7	56	110.5	65.6	102.5
Screen Magnifier	86.8	53.6	70	114.2	44.8	104.5
Total	79.7	61.7	59	112	56.7	102.5
Perceived task success (in %)						
	Level NA			Level AA		
	Mean	SD	Median	Mean	SD	Median
Screen Reader	94.4	13	100	94.1	17.6	100
Screen Magnifier	90.9	30.2	100	93.9	13.5	100
Total	92.8	22.4	100	94	15.9	100
Task satisfaction (ASQ, 1-7)						
	Level NA			Level AA		
	Mean	SD	Median	Mean	SD	Median
Screen Reader	5.1	2	6	4.9	2.1	6
Screen Magnifier	4.4	2.6	6	5.1	2.2	6.3
Total	4.8	2.3	6	5	2.1	6
Perceived usability (WAMMI, 1-7)						
	Level NA			Level AA		
	Mean	SD	Median	Mean	SD	Median
Screen Reader	5.5	0.9	5.6	5.3	1.4	6
Screen Magnifier	5.6	1.6	6.4	5.5	1.6	6
Total	5.5	1.2	5.8	5.4	1.4	6
Visual aesthetic (VisAWI, 1-7)						
	Level NA			Level AA		
	Mean	SD	Median	Mean	SD	Median
Screen Reader	6	0.6	6	6.1	0.9	6.2
Screen Magnifier	6	1.2	6.5	5.7	1.5	6
Total	6	0.9	6	5.9	1.2	6.2
Pragmatic quality (AttrakDiff, 1-7)						
	Level NA			Level AA		
	Mean	SD	Median	Mean	SD	Median
Screen Reader	5.6	0.8	5.6	5.6	1.1	6
Screen Magnifier	5.8	1.9	6.8	6	1.2	6.8
Total	5.7	1.4	6	5.8	1.1	6
Hedonic quality (AttrakDiff, 1-7)						
	Level NA			Level AA		
	Mean	SD	Median	Mean	SD	Median
Screen Reader	4.8	1	4.9	5	1.2	4.5
Screen Magnifier	5.5	1.7	6.5	5.5	1.2	5.2
Total	5.1	1.4	5.2	5.2	1.2	5
Likelihood to recommend (NPS, total score of -100% to +100%)						
	Level NA			Level AA		
Screen Reader	25			17.7		
Screen Magnifier	9.1			18.2		
Total	17.4			17.9		

[1] Only successful attempts were used to compare task completion times.
[2] Fully blind participants were not asked for their ratings regarding the visual aesthetic.

user group ($F(1, 116) = 1.51$, $p = .221$, partial $\eta^2 = .013$, 90% CI[0.00, 0.07]).

Perceived task success. The perception of task success did not differ significantly for conformance ($F(1, 127) = 0.55$, $p = .459$, partial $\eta^2 = .004$, 90% CI[0.00, 0.04]) or the user group ($F(1, 127) = 3.53$, $p = .062$, partial $\eta^2 = .027$, 90% CI[0.00, 0.09]), and no significant interaction between conformance and the user group ($F(1, 127) = 1.94$, $p = .166$, partial $\eta^2 = .015$, 90% CI[0.00, 0.07]) was found.

Task satisfaction. Task satisfaction was measured with the After Scenario Questionnaire (ASQ). We found no significant differences in task satisfaction as a function of conformance to web accessibility standards ($F(1, 127) = 0.65$, $p = .421$, partial $\eta^2 = .005$, 90% CI[0.00,

0.05]). There were also no significant differences for the user group ($F(1, 127) = 0.23$, $p = .631$, partial $\eta^2 = .002$, 90% CI[0.00, 0.03]) or for the interaction between conformance and the user group ($F(1, 127) = 0.00$, $p = .987$, partial $\eta^2 < .001$, 90% CI[0.00, 0.00]).

Perceived usability. Perceived usability was measured with the Website Analysis and MeasureMent Inventory (WAMMI; $\alpha = .95$, $\omega_u = .96$, 95% CI[0.94,0.97]). The ratings for perceived usability did not differ significantly for conformance ($F(1, 127) = 0.46$, $p = .500$, partial $\eta^2 = .004$, 90% CI[0.00, 0.04]) or for the user group ($F(1, 127) = 0.10$, $p = .758$, partial $\eta^2 = .001$, 90% CI[0.00, 0.02]), and no significant interaction between conformance and the user group ($F(1, 127) = 0.53$, $p = .470$, partial $\eta^2 = .004$, 90% CI[0.00, 0.04]) was found.

Visual aesthetic. The visual aesthetic was measured with the Visual Aesthetics of Websites Inventory (VisAWI; $\alpha = .94$, $\omega_u = .94$, 95% CI[0.92, 0.97]). Fully blind participants were not asked for their rating regarding the visual aesthetic. We did not find a significant difference for the visual aesthetic as a function of conformance ($F(1, 121) = 1.67$, $p = .199$, partial $\eta^2 = .014$, 90% CI[0.00, 0.07]), of the user group ($F(1, 121) = 3.86$, $p = .052$, partial $\eta^2 = .031$, 90% CI[0.00, 0.10]), or of the interaction between conformance and the user group ($F(1, 121) = 3.03$, $p = .084$, partial $\eta^2 = .024$, 90% CI[0.00,0.09]).

Pragmatic quality. The pragmatic quality was measured with the pragmatic quality scale of the AttrakDiff2 ($\alpha = .89$, $\omega_u = .89$, 95% CI[0.85, 0.93]). No significant differences were found for pragmatic quality as a function of conformance ($F(1, 127) = 0.40$, $p = .527$, partial $\eta^2 = .003$, 90% CI[0.00, 0.04]), as a function of the user group ($F(1, 127) = 0.03$, $p = .868$, partial $\eta^2 < .001$, 90% CI[0.00, 0.02]), or as a function of the interaction between conformance and the user group ($F(1, 127) = 1.05$, $p = .308$, partial $\eta^2 = .008$, 90% CI[0.00, 0.05]).

Hedonic quality. The hedonic quality was measured with the hedonic-quality scale of the AttrakDiff2 ($\alpha = .89$, $\omega_u = .89$, 95% CI[0.87, 0.93]). As for its pragmatic counterpart, no significant differences were found for the hedonic quality as a function of conformance ($F(1, 127) = 0.85$, $p = .359$, partial $\eta^2 = .007$, 90% CI[0.00, 0.05]), the user group ($F(1, 127) = 3.33$, $p = .071$, partial $\eta^2 = .026$, 90% CI[0.00, 0.09]), or the interaction between conformance and the user group ($F(1, 127) = 2.09$, $p = .150$, partial $\eta^2 = .016$, 90% CI[0.00, 0.07]).

Likelihood to recommend. Participants' ratings for the likelihood that they would recommend the online shop showed no significant differences for conformance ($F(1, 127) = 0.08$, $p = .772$, partial $\eta^2 = .001$, 90% CI[0.00, 0.02]), the user group ($F(1, 127) = 3.25$, $p = .074$, partial $\eta^2 = .025$, 90% CI[0.00, 0.09]), or the interaction between conformance and the user group ($F(1, 127) = 0.81$, $p = .369$, partial $\eta^2 = .006$, 90% CI[0.00, 0.05]).

4.3. Qualitative measures

In the analysis of the open-ended answers, a total number of 426 codes were assigned. Solely related to the study materials itself were 82 codes, which were subsequently removed from the dataset. A total number of 344 (N positive = 160, 46.5%; N negative = 184, 53.5%) codes were included in the main analysis. They were related to web accessibility (N = 36, 10.5%), usability (227, 66%), user experience (56, 16%), or coded as not assignable (25, 7.5%). The results of the analysis are presented in Table 6.

Due to the small number of codes in some categories, only the total numbers of positive and negative codes were statistically analysed. Significantly more positive experiences (χ^2 (1, N = 105) = 10.37, $p < .01$) were reported by participants with visual impairments in the AA condition compared to the NA condition, whereas no significant difference was found in negative experiences (χ^2 (1, N = 81) = 2.78, $p = .096$). Participants without visual impairments reported significantly fewer positive experiences (χ^2 (1, $N = 55$) = 6.56, $p < .05$), but also significantly fewer negative experiences (χ^2 (1, N = 103) = 7.08, $p < .01$) for the AA condition compared to the NA condition.

5. Discussion

5.1. Effects of compliance

Our motivation for the present study was to deepen our understanding of the relationship between compliance with web accessibility standards and measures related to usability and user experience. The experimental setup allowed us to manipulate the conformance of an online shop to a compliant and non-compliant level. Participants with and without visual impairments took part in the study. For the measures related to usability, we found no statistically significant differences between the compliant and non-compliant versions for both user groups. This was the case for the performance measures (task completion rate, task completion time), the outcomes related to usability (perceived task success, task satisfaction, perceived usability, pragmatic quality), and the measures related to user experience (visual aesthetic, hedonic quality). Small effect sizes were found for all the outcomes as a function of conformance to web accessibility standards (range between a partial η^2 of 0.001 for task completion rate up to 0.016 for task completion time). However, the analysis of the open-ended answers showed significantly more positive experiences for participants with visual impairments, which is as expected especially apparent for experiences related to web accessibility. In contrast, participants without visual impairments reported significantly fewer positive experiences but also significantly fewer negative experiences in the compliant condition than in the non-compliant condition, as well as overall. Given that negative user experiences have a stronger impact on the overall experience (Vaish et al., 2008), we consider this a benefit of the version that was compliant with web accessibility standards.

The present findings reveal a complex relationship between compliance with web accessibility standards and outcomes related to usability and user experience. They contrast with the positive effects found in the series of studies by Schmutz et al. (2016, 2017b,a) as well as in the studies by Pascual et al. (2014b,a, 2015), where compliance with standards significantly increased all the comparable measures. Further, although a similar experimental setup was used, substantially smaller effect sizes were observed in the present study (e.g., compared to Schmutz et al., 2017a; partial η^2 for perceived usability of 0.004 vs 0.08; partial η^2 for the visual aesthetic of 0.014 vs 0.07). These differences potentially arise from the users that participated in the study and the type of website they interacted with. Both factors indicate that the present findings demand a more differentiated perspective on conformance-based approaches to web accessibility.

Whereas the previous series of studies conducted in a laboratory or moderated remote setting, participation in the present study was fully online and unmoderated. This meant that solid skills in using the web and assistive technologies were required to participate in the first place. This is also reflected in the high self-rated experience with computers in general and with websites in particular. Since users with assistive technologies show a steep learning curve and often establish elaborate tactics to overcome web accessibility issues (Bigham et al., 2017; Moreno et al., 2020; Vigo and Harper, 2014, 2013), compliance with standards may have had less importance in the present setting. Previous studies have discussed whether participants with and without visual impairments encounter similar problems but perceive them differently in their severity (Petrie and Kheir, 2007). A comparable pattern could exist between more and less experienced users. Another main difference between the present study and past studies were the materials used. An online shop with common, highly standardised interaction patterns may have made it easier for the participants to circumvent web accessibility hurdles than in studies with less familiar materials. For example, to complete a purchase on an online shop, most users will look for some kind of checkout button, usually identified by a shopping cart. This pattern is so common that such a button is probably even found when a label or an alternative text is not optimally implemented. This illustrates that the actual impact of a barrier can be

Table 6
Number of codes per category assigned to the open-ended answers.

	Impaired eyesight		Unimpaired eyesight		Total	
	Level NA N = 84 (%)	Level AA N = 102 (%)	Level NA N = 102 (%)	Level AA N = 56 (%)	Level NA N = 186 (%)	Level AA N = 158 (%)
Positive	36 (34)	69 (66)	37 (67)	18 (33)	73 (46)	87 (54)
Web accessibility	2 (14)	12 (86)	2 (67)	1 (33)	4 (24)	13 (76)
Usability	15 (35)	28 (65)	23 (64)	13 (36)	38 (48)	41 (52)
User experience	10 (32)	21 (68)	6 (75)	2 (25)	16 (41)	23 (59)
Not assignable	9 (53)	8 (47)	6 (75)	2 (25)	15 (60)	10 (40)
Negative	48 (59)	33 (41)	65 (63)	38 (37)	113 (61)	71 (39)
Web accessibility	7 (54)	6 (46)	6 (100)	–	13 (68)	6 (32)
Usability	36 (61)	23 (39)	54 (61)	35 (39)	90 (61)	58 (39)
User experience	5 (56)	4 (44)	5 (62)	3 (38)	10 (59)	7 (41)
Not assignable	–	–	–	–	–	–

hard to determine and its perceived severity depends on the specific context (Abou-Zahra, 2008).

These considerations are in line with previous discussions that conformance to standards should be seen as a first step in web accessibility (Power et al., 2012). The observed variation of the effect sizes between the studies indicates that the impact of compliance may vary from product to product. Since every digital technology has its own distinct users and context, what is covered by standards and the perceived severity of barriers can also differ (Aizpurua et al., 2015). A significant effect of compliance with web accessibility standards alone on outcomes related to usability and user experience outcomes should therefore not be expected in every case. But this does not mean that conformance to standards is useless and not worth the effort. This is reflected in the open-ended answers of the present study, in which users with visual impairments reported more positive experiences and users without visual impairments reported fewer negative experiences in the compliant condition. Efforts to achieve compliance with web accessibility standards are a good starting point and an important milestone for accessible information and services (Power et al., 2018). To truly support inclusion, it is essential to go further and to complement conformance-based approaches with other methodologies.

5.2. Complementing compliance

The present findings highlight that inclusive experiences can only be designed with a thorough understanding of the user needs in a specific context. They also show that the checklist-based approach, as it is applied in WCAG 2.0, is reaching its limits. This can be observed in the open-ended comments from participants with visual impairments, where only a fraction of the experiences (N = 36, 10.5%) were classified as related to web accessibility. It is therefore debatable whether the participants encountered problems regarding web accessibility that are not sufficiently addressed in the given standards or whether the problems are tied to a specific context and go beyond what should be part of common guidelines. A set of standard criteria can ensure that a product is in line with important web accessibility requirements, but it cannot account for users' actual experiences in a given context (Horton et al., 2015).

Although the extension of conformance-based methodologies is not a new demand, only few tools and techniques have been established for capturing and understanding the digital experiences of users with disabilities (Power et al., 2018). The present results highlight the need for a discussion of approaches that may improve the outcomes related to usability and user experience for all user groups. Two propositions are the use of frameworks or principles and the application of user-centred or participatory design methodologies. In contrast to common web accessibility checklists, inclusive design frameworks or principles have a broader scope and try to guide the design process rather than pointing out technical implementations. Two examples for such frameworks are the Inclusive Design Principles (Swan et al., 2017) and the Accessible User Experience Principles (Horton and Quesenbery, 2014). The former proposes seven broad conventions (e.g., consider situation, be consistent, give control), and the latter proposes nine (e.g., people first, clear purpose, solid structure). Another approach is to build inclusive experiences based on established methodologies from the related field of usability and user experience: user-centred design and participatory design methods offer opportunities for better engaging users with disabilities (Henry, 2007). Whereas traditional user-centred design methods involve users but leave design decisions to professionals (e.g., contextual enquiries, user interviews, usability testings), participatory design invites users to actively define and shape the design of a product (Oswal, 2019). Standards, principles, and user involvement should complement each other to acquire a thorough understanding of users' perspectives and to facilitate the design of inclusive experiences (e.g., Brewster et al., 2019; Vollenwyder et al., 2020a).

The discussed results support current industry efforts to establish a more comprehensive view on web accessibility. For example, an early draft of WCAG 3.0 contains considerations such as a more differentiated scoring mechanism with a range from very poor to excellent for each criterion and the demand for holistic tests (Spellman et al., 2021). While further details are not yet available, the outline explicitly names user-centred design methods as part of such requirements. It is hard to predict how these first ideas will evolve and how they will be received in practice. In principle, the present results support the directions outlined in WCAG 3.0 as a reasonable advancement.

5.3. Limitations and future research

The present study has several limitations. First, the selection of the study materials could be improved for future studies. Creating an online shop from scratch had its benefits because it allowed high methodological rigour with very specific manipulations according to WCAG 2.0. But it is possible that participants perceived the content and manipulations as rather artificial. Using versions of an existing website, ideally from before and after it is optimised according to WCAG 2.0, would better account for the complexity of web accessibility issues and allow a higher external validity in a future study. This would also address the issue that a considerable percentage of users in all experimental conditions failed with some of the given tasks. One possible reason could be that the basic usability of the online shop used in the study was not optimal. The effect of a better compliance to web accessibility standards may have been diminished by such problems. Further, study materials from a less familiar context than online shopping should be considered. Based on the present results that compliance with standards alone does not necessarily provide improved usability and a satisfying user experience, it would also be interesting to compare the outcomes of websites that did and did not undergo an inclusive design process. User-centred methodologies, as outlined in the preceding section, could be applied in future studies to acquire insights about the most promising methods for creating inclusive experiences.

Second, as outlined in the main results, the study was conducted in a fully remote and unmoderated manner. As discussed, effects such as

a self-selection bias among more experienced users or more dropouts among less experienced users cannot be ruled out. Future studies should take this into account and also include users that are less proficient with using the web. Further, it was not possible to observe the participants or for them to give immediate feedback, which makes it more challenging to capture a comprehensive picture of their experiences during the study. However, this setup allowed an equal or larger sample size compared to previous studies in the field (e.g., Schmutz et al., 2017b; Pascual et al., 2014b). Additionally, it allowed the participants to take part in their familiar surroundings and with their individual setup of assistive technologies (Miao et al., 2014). A replication of this study, for instance in a laboratory setting, would therefore imply various other limitations. A more promising option for future research could be to conduct entire studies or selected parts of them in synchronous remote sessions (Schmutz et al., 2017b) or to ask participants to record and comment on their attempts. Both approaches would allow a richer picture of experiences with digital technologies.

Finally, the present study equates web accessibility with efforts to make the web accessible to users with visual impairments and ignores users with other disabilities, a pattern which is frequently found in research and practice (Vollenwyder et al., 2020b). Although users with visual impairments are an important target group for web accessibility, it is essential to highlight that other sensory, motor, or cognitive impairments should receive equal attention. Replicating this study with another group of users with disabilities would benefit the validity of the results and provide more comprehensive insights into experiences with digital technologies. Further, the question regarding an adequate representation of various user groups leads to another research gap that future studies could address. The recommended application of user-centred design methodologies also requires good strategies for recruiting users with disabilities (Henry, 2007). In the context of web accessibility, this is a challenging task since a category such as "motor impairment" is vast and includes a multitude of disabilities that can substantially influence a user's perception of interactions with digital technologies. Best practices for approaching this complexity would therefore be helpful in research and practice.

5.4. Implications for practitioners

In practice, web professionals often face the challenge of advocating web accessibility efforts towards their stakeholders (Vollenwyder et al., 2019). The present results provide arguments in favour of taking a differentiated position in such debates. First of all, it remains vital to consider existing standards such as WCAG 2.0 while designing accessible information and services on the web. Even large projects will not be able to actively include users that represent the diversity of disabilities, assistive technologies, and levels of experience with digital technologies (Henry, 2007). Conformance to web accessibility standards can provide a solid foundation that covers the most important technical aspects. At the same time, it is also important to emphasise that significant benefits cannot be expected from compliance in every case. For example, stakeholders may be disappointed if they still receive negative user feedback for their freshly certified website. As a web professional, it is important to provide context, to explain how such feedback can be interpreted and why it does not question previous web accessibility efforts. As seen in the present study, it might be necessary to explain that a product in a highly standardised context, such as an online shop, does not stand out through compliance with standards alone.

Conformance to web accessibility standards should be seen as a first step for inclusive experiences. Additionally, we recommended advocating for a user-centred perspective during the whole design process of a product and including the user's perspective as early as possible. For example, users with disabilities and with various proficiency levels should be included in user research activities ahead of a project, design drafts should be challenged before being handed over to developers, and testings with users should be scheduled soon after the first implementation. Such measures are usually possible with limited resources, and they prevent expensive retrofitting activities in a later stage of product design (Urban and Burks, 2006). Further, the involvement of users with disabilities has proven to be an effective method for fostering awareness and knowledge about web accessibility at all organisational levels (Vollenwyder et al., 2020a).

Finally, we encourage all web professionals to share their inclusive design examples. Since there is still a lack of established tools and techniques for capturing and understanding the digital experiences of users with disabilities, such insights are valuable for developing best practices to complement conformance-based approaches. Additionally, emphasising personal responsibility in every professional role is an important motivation for considering web accessibility (Vollenwyder et al., 2019). Intensive collaboration among the community of web professionals supports this notion and provides opportunities to reach out for help during an inclusive design process.

6. Conclusion

The present work investigated how compliance with web accessibility standards shapes the experiences of users with and without disabilities. The results from a randomised controlled experiment showed that significant effects on outcomes related to usability and user experience cannot be expected for all users in every context. Since daily work on accessible information and services often relies exclusively on given standards, we recommend adopting a more differentiated perspective on what can be achieved through conformance. While standards serve as a solid foundation, we propose complementing them with user-centred and participatory design approaches. We hope our research encourages web professionals to continue their quest towards a web for everyone.

CRediT authorship contribution statement

Beat Vollenwyder: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **Serge Petralito:** Conceptualization, Methodology, Software. **Glena H. Iten:** Conceptualization, Methodology, Software. **Florian Brühlmann:** Conceptualization, Methodology, Formal analysis, Writing – review & editing. **Klaus Opwis:** Supervision. **Elisa D. Mekler:** Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We would like to thank the Swiss Federation of the Blind and Visually Impaired (SBV), the foundation Access for all, and TestingTime for their help in carrying out the research. We are grateful to Andrea Schneider, Sven Schmutz, and Jurek Kirakowski for their helpful inputs and to Anthony Mahler for his editorial assistance. The study was registered with the Institutional Review Board of the first author's university under the number D-007-17.

Appendix. Manipulations

See Table A.7.

Table A.7
Overview of the manipulated website characteristics according to WCAG 2.0.

Number	WCAG name	NA implementation	AA implementation
Perceivable			
1.1.1	Non-text Content	Text alternatives were not provided for images and icons.	Every image and icon on the website provided an appropriate text alternative
1.3.1	Info and Relationships	Content was not fully structured by heading tags. Forms did not contain label tags. Tables did not contain heading tags.	Content was fully structured by heading tags. Forms contain label tags. Tables contain heading tags.
1.3.2	Meaningful Sequence	Content sequence for the page "Information" was visually but not programmatically meaningful.	Content sequence for the page "Information" was visually and programmatically meaningful.
1.4.1	Use of Colour	Active navigation and links were indicated by a different colour.	Active navigation and links were indicated by underlined text.
1.4.3	Contrast (Minimum)	The contrast between headings and background was 2.9:1 (#989898 | #FFFFFF). The contrast between text and background was 4.1:1 (#7D7D7D | #FFFFFF). The contrast between text and buttons was 2.2:1 (#FFFFFF | #43C94D). Background images in the heading section were not adjusted for good readability.	The contrast between headings and background as well as text and background was 15.9:1 (#222222 | #FFFFFF). The contrast between text and buttons was 4.5:1 (#FFFFFF | #128A1B). Background images in the heading section were adjusted for good readability.
1.4.8	Visual Presentation	Text blocks did not have a maximum width and were justified.	Text blocks had a maximum width of 80 characters and were left aligned.
Operable			
2.1.1	Keyboard	Autocomplete was not operable with a keyboard.	Autocomplete was fully operable with a keyboard.
2.4.1	Bypass block	Skip links and access keys were not available.	Skip links and access keys were available; they were visually hidden at the top of the website.
2.4.2	Page Titled	Page titles were not adapted to the content.	Page titles were adapted to the content.
2.4.3	Focus Order	Using the tab key on product pages and in the contact form did not move one between the fields in a meaningful order.	Using the tab key moved one between all the fields received focus in a meaningful order.
2.4.5	Multiple Ways	An alternative way to the products was not provided.	Direct links in the footer section provided an alternative way to the products.
2.4.6	Headings and Labels	Some headings were shortened to be less descriptive.	Headings were descriptive for the related content.
2.4.7	Focus Visible	Keyboard focus indicator was not visible.	Keyboard focus indicator was visible and optimised for comfortable recognition.
2.4.10	Section Headings	Some section headings were removed.	Section headings were used to organise the content.
Understandable			
3.1.1	Language of Page	Default language was fixed to English.	Default language matched the current language.
3.1.4	Abbreviations	Product origins were declared with abbreviations of the countries.	Product origins were declared with full names of the countries.
3.2.3	Consistent Navigation	Not all navigation links were presented on the search and checkout page.	Navigational links appeared consistently.
3.2.4	Consistent Identification	Some buttons to add products to the cart differed in design, although they had the same functionality.	Buttons were designed consistently.
3.3.1	Error Identification	Error identification was not provided for forms.	Error identification using "required" attributes was provided for all forms.

(*continued on next page*)

Table A.7 (*continued*).

Number	WCAG name	NA implementation	AA implementation
3.3.2	Labels or Instructions	Required fields were presented in bold text without additional text alternatives.	Required fields were presented in bold text, were labelled with an asterisk, and were accompanied by an additional hidden text alternative.
Robust			
4.1.2	Name, Role, Value	Role attributes were not used.	Role attributes were used for alert messages and for the validation messages in forms.

References

Abou-Zahra, S., 2008. Web accessibility evaluation. In: Harper, S., Yesilada, Y. (Eds.), Web Accessibility: A Foundation for Research. In: Human-Computer Interaction Series, Springer London, pp. 79–106. http://dx.doi.org/10.1007/978-1-84800-050-6_7.

Aizpurua, A., Arrue, M., Vigo, M., 2015. Prejudices, memories, expectations and confidence influence experienced accessibility on the Web. Comput. Hum. Behav. 51, 152–160. http://dx.doi.org/10.1016/j.chb.2015.04.035.

Aizpurua, A., Harper, S., Vigo, M., 2016. Exploring the relationship between web accessibility and user experience. Int. J. Hum.-Comput. Stud. 91, 13–23. http://dx.doi.org/10.1016/j.ijhcs.2016.03.008.

Bendixen, K., Benktzon, M., 2015. Design for All in Scandinavia – A strong concept. Applied Ergon. 46, 248–257. http://dx.doi.org/10.1016/j.apergo.2013.03.004.

Bigham, J.P., Lin, I., Savage, S., 2017. The effects of "not knowing what you don't know" on web accessibility for blind web users. In: ASSETS '17: Proceedings of the 19th International ACM SIGACCESS Conference on Computers and Accessibility. pp. 101–109. http://dx.doi.org/10.1145/3132525.3132533.

Brajnik, G., Yesilada, Y., Harper, S., 2011. The expertise effect on web accessibility evaluation methods. Hum.-Comput. Interact. 26 (3), 246–283. http://dx.doi.org/10.1080/07370024.2011.601670.

Brajnik, G., Yesilada, Y., Harper, S., 2012. Is accessibility conformance an elusive property? A study of validity and reliability of WCAG 2.0. ACM Trans. Access. Comput. (TACCESS) 4 (2), 8–28. http://dx.doi.org/10.1145/2141943.2141946.

Brewster, S., Fitzpatrick, G., Cox, A., Kostakos, V., Metatla, O., Oldfield, A., Ahmed, T., Vafeas, A., Miglani, S., 2019. Voice user interfaces in schools. In: Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems. pp. 1–15. http://dx.doi.org/10.1145/3290605.3300608.

Brühlmann, F., Petralito, S., Aeschbach, L.F., Opwis, K., 2020. The quality of data collected online: An investigation of careless responding in a crowdsourced sample. Methods Psychol. 2, 100022. http://dx.doi.org/10.1016/j.metip.2020.100022.

Caldwell, B., Cooper, M., Reid, L.G., Vanderheiden, G., 2008. Web content accessibility guidelines (WCAG) 2.0. URL: http://www.w3.org/TR/2008/REC-WCAG20-20081211.

Chisholm, W., Vanderheiden, G., Jacobs, I., 1999. Web content accessibility guidelines 1.0. URL: https://www.w3.org/TR/WCAG10.

Clarkson, P.J., Coleman, R., Keates, S., Lebbon, C., 2003. Inclusive Design: Design for the Whole Population, http://dx.doi.org/10.1007/978-1-4471-0001-0.

Cooper, M., Sloan, D., Kelly, B., Lewthwaite, S., 2012. A challenge to web accessibility metrics and guidelines: putting people and processes first. In: W4A '12: Proceedings of the International Cross-Disciplinary Conference on Web Accessibility. ACM, pp. 1–4. http://dx.doi.org/10.1145/2207016.2207028.

Curran, P.G., 2016. Methods for the detection of carelessly invalid responses in survey data. J. Exp. Soc. Psychol. 66, 4–19. http://dx.doi.org/10.1016/j.jesp.2015.07.006.

Ellcessor, E., 2014. <ALT="Textbooks">: Web accessibility myths as negotiated industrial lore. Crit. Stud. Media Commun. 31 (5), 448–463. http://dx.doi.org/10.1080/15295036.2014.919660.

Fajardo, I., Ávila, V., Ferrer, A., Tavares, G., Gómez, M., Hernández, A., 2013. Easy-to-read texts for students with intellectual disability: Linguistic factors affecting comprehension. J. Appl. Res. Intell. Disabilities 27 (3), 212–225. http://dx.doi.org/10.1111/jar.12065.

Flick, U., Kardoff, E.v., Steinke, I., 2004. A Companion to Qualitative Research. Sage.

Harper, S., Yesilada, Y., 2008. Web accessibility and guidelines. In: Harper, S., Yesilada, Y. (Eds.), Web Accessibility: A Foundation for Research. In: Web Accessibility, Springer London, pp. 61–78. http://dx.doi.org/10.1007/978-1-84800-050-6_6.

Hassenzahl, M., Diefenbach, S., Göritz, A., 2010. Needs, affect, and interactive products – Facets of user experience. Interact. Comput. 22 (5), 353–362. http://dx.doi.org/10.1016/j.intcom.2010.04.002.

Hassenzahl, M., Monk, A., 2010. The inference of perceived usability from beauty. Hum.-Comput. Interact. 25, 235–260. http://dx.doi.org/10.1080/07370024201050013 9.

Heim, M., Uebelbacher, A., Winkelmann-Ackermann, S., 2020. Schweizer Accessibility-Studie Onlineshops: Eine Studie Von "Zugang Für Alle" Zur Barrierefreiheit Von Onlineshops in Der Schweiz Im Jahr 2020.

Henry, S.L., 2006. Understanding web accessibility. In: Web Accessibility: Web Standards and Regulatory Compliance. A Press, pp. 1–51. http://dx.doi.org/10.1007/978-1-4302-0188-5_1.

Henry, S.L., 2007. Just Ask: Integrating Accessibility Throughout Design. Lulu.com, URL: http://uiaccess.com/accessucd/contents.html.

Holliday, E.L., 2020. The compliance mindset. In: The 22nd International ACM SIGACCESS Conference on Computers and Accessibility. Vol. 134, pp. 1–3. http://dx.doi.org/10.1145/3373625.3417078.

Honaker, J., King, G., Blackwell, M., 2021. Amelia: A program for missing data. URL: https://CRAN.R-project.org/package=Amelia.

Horton, S., Quesenbery, W., 2014. A Web for Everyone. Rosenfeld Media, URL: http://rosenfeldmedia.com/books/a-web-for-everyone/.

Horton, S., Sloan, D., Swan, H., 2015. Complementing Standards By Demonstrating Commitment and Progress. In: The 12th Web for All Conference, ACM, http://dx.doi.org/10.1145/2745555.2746654.

International Organization for Standardization, 2019. Ergonomics of human-system interaction — Part 210: Human-centred design for interactive systems (ISO 9241-210:2019). URL: https://www.iso.org/standard/77520.html.

Iwarsson, S., Ståhl, A., 2009. Accessibility, usability and universal design—positioning and definition of concepts describing person-environment relationships. Disabil. Rehabil. 25 (2), 57–66. http://dx.doi.org/10.1080/dre.25.2.57.66.

Kalbag, L., 2017. Accessibility for Everyone. A Book Apart.

Karreman, J., Geest, T.v.d., Buursink, E., 2007. Accessible website content guidelines for users with intellectual disabilities. J. Appl. Res. Intell. Disabilities 20 (6), 510–518. http://dx.doi.org/10.1111/j.1468-3148.2006.00353.x.

Kirakowski, J., Claridge, N., Whitehand, R., 1998. Human centered measures of success in web site design.

Landis, J.R., Koch, G.G., 1977. The measurement of observer agreement for categorical data. Biometrics 33 (1), 159–174.

Lewis, J.R., 1995. IBM computer usability satisfaction questionnaires: Psychometric evaluation and instructions for use. Int. J. Hum.-Comput. Interact. 7 (1), 57–78. http://dx.doi.org/10.1080/10447319509526110.

Mbipom, G., Harper, S., 2011. The Interplay Between Web Aesthetics and Accessibility. In: The proceedings of the 13th international ACM SIGACCESS conference, ACM, http://dx.doi.org/10.1145/2049536.2049564.

Miao, M., Pham, H.A., Friebe, J., Weber, G., 2014. Contrasting usability evaluation methods with blind users. Univers. Access Inf. Soc. 15 (1), 63–76. http://dx.doi.org/10.1007/s10209-014-0378-8.

Moreno, L., Valencia, X., Pérez, J.E., Arrue, M., 2020. An exploratory study of web adaptation techniques for people with low vision. Univers. Access Inf. Soc. 84 (3), 13. http://dx.doi.org/10.1007/s10209-020-00727-6.

Moshagen, M., Thielsch, M., 2013. A short version of the visual aesthetics of websites inventory. Behav. Inf. Technol. 32 (12), 1305–1311. http://dx.doi.org/10.1080/0144929x.2012.694910.

Oswal, S.K., 2019. Breaking the exclusionary boundary between user experience and access: steps toward making UX inclusive of users with disabilities. In: Proceedings of the 37th ACM International Conference on the Design of Communication. pp. 1–8. http://dx.doi.org/10.1145/3328020.3353957.

Pascual, A., Ribera, M., Granollers, T., 2014a. Impact of web accessibility barriers on users with hearing impairment. In: Interacción '14: Proceedings of the XV International Conference on Human Computer Interaction. pp. 1–2. http://dx.doi.org/10.1145/2662253.2662261.

Pascual, A., Ribera, M., Granollers, T., 2015. Impact of accessibility barriers on the mood of users with motor and dexterity impairments. J. Access. Des. All 1–27. http://dx.doi.org/10.17411/jacces.v5i1.93.

Pascual, A., Ribera, M., Granollers, T., Coiduras, J.L., 2014b. Impact of accessibility barriers on the mood of blind, low-vision and sighted users. Proc. - Proc. Comput. Sci. 27, 431–440. http://dx.doi.org/10.1016/j.procs.2014.02.047.

Pereira, L.S., Archambault, D., 2018. Correlating navigation barriers on web 2.0 with accessibility guidelines. In: Miesenberger, K., Kouroupetroglou, G. (Eds.), Computers Helping People with Special Needs. ICCHP 2018. Lecture Notes in Computer Science. Vol. 10896, Springer Berlin Heidelberg, pp. 13–21. http://dx.doi.org/10.1007/978-3-319-94277-3_3.

Petrie, H., Hamilton, F., King, N., 2004. Tension, what tension? Website accessibility and visual design. In: W4A '04: Proceedings of the 2004 International Cross-Disciplinary Workshop on Web Accessibility (W4A). In: Proceedings of the international cross-disciplinary workshop on Web accessibility (W4A), pp. 13–18. http://dx.doi.org/10.1145/990657.990660.

Petrie, H., Kheir, O., 2007. The relationship between accessibility and usability of websites. In: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems. pp. 397–406. http://dx.doi.org/10.1145/1240624.1240688.

Petrie, H., Savva, A., Power, C., 2015. Towards a unified definition of web accessibility. In: The 12th Web for All Conference. pp. 1–13. http://dx.doi.org/10.1145/2745555.2746653.

Power, C., Cairns, P., Barlet, M., 2018. Inclusion in the third wave: Access to experience. In: Filimowicz, M., Tzankova, V. (Eds.), New Directions in Third Wave Human-Computer Interaction: Volume 1 - Technologies. In: Human–Computer Interaction Series, Vol. 91, Springer London, pp. 163–181. http://dx.doi.org/10.1007/978-3-319-73356-2_10.

Power, C., Freire, A., Petrie, H., Swallow, D., 2012. Guidelines are only half of the story: accessibility problems encountered by blind users on the web. In: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems. pp. 433–442. http://dx.doi.org/10.1145/2207676.2207736.

Reichheld, F.F., 2003. The one number you need to grow. Harv. Bus. Rev. 81 (12), 46–54.

Rømen, D., Svanæs, D., 2011. Validating WCAG versions 1.0 and 2.0 through usability testing with disabled users. Univers. Access Inf. Soc. 11 (4), 375–385. http://dx.doi.org/10.1007/s10209-011-0259-3.

Sauer, J., Sonderegger, A., Schmutz, S., 2020. Usability, user experience and accessibility: towards an integrative model. Ergonomics 63 (10), 1–23. http://dx.doi.org/10.1080/00140139.2020.1774080.

Schmutz, S., Sonderegger, A., Sauer, J., 2016. Implementing recommendations from web accessibility guidelines: Would they also provide benefits to nondisabled users. Hum. Factors: J. Hum. Factors Ergon. Soc. 58 (4), 611–629. http://dx.doi.org/10.1177/0018720816640962.

Schmutz, S., Sonderegger, A., Sauer, J., 2017a. Effects of accessible website design on nondisabled users: age and device as moderating factors. Ergonomics 47 (3), 1–13. http://dx.doi.org/10.1080/00140139.2017.1405080.

Schmutz, S., Sonderegger, A., Sauer, J., 2017b. Implementing recommendations from web accessibility guidelines: A comparative study of nondisabled users and users with visual impairments. Hum. Factors: J. Hum. Factors Ergon. Soc. 59 (6), 956–972. http://dx.doi.org/10.1177/0018720817708397.

Schmutz, S., Sonderegger, A., Sauer, J., 2019. Easy-to-read language in disability-friendly web sites: Effects on nondisabled users. Applied Ergon. 74, 97–106. http://dx.doi.org/10.1016/j.apergo.2018.08.013.

Shneiderman, B., 2000. Universal usability. Commun. ACM 43 (5), 84–91. http://dx.doi.org/10.1145/332833.332843.

Spellman, J., Montgomery, R.B., Lauriat, S., Cooper, M., 2021. W3C accessibility guidelines (WCAG) 3.0. URL: https://www.w3.org/TR/wcag-3.0/.

Swan, H., Pouncey, I., Pickering, H., Watson, L., 2017. Inclusive design principles. URL: https://inclusivedesignprinciples.org/.

Tierney, N., aut, D., McBain, M., Fay, C., O'Hara-Wild, M., Hester, J., Smith, L., Heiss, A., 2021. naniar: Data Structures, Summaries, and Visualisations for Missing Data. URL: https://CRAN.R-project.org/package=naniar.

Urban, M., Burks, M.R., 2006. Implementing accessibility in the enterprise. In: Web Accessibility: Web Standards and Regulatory Compliance. A Press, pp. 69–83. http://dx.doi.org/10.1007/978-1-4302-0188-5_3.

Vaish, A., Grossmann, T., Woodward, A., 2008. Not all emotions are created equal: The negativity bias in social–emotional development. Psychol. Bull. 134 (3), 383–403. http://dx.doi.org/10.1037/0033-2909.134.3.383.

Vigo, M., Brown, J., Conway, V., 2013. Benchmarking Web Accessibility Evaluation Tools: Measuring the Harm of Sole Reliance on Automated Tests. In: The 10th International Cross-Disciplinary Conference, ACM, http://dx.doi.org/10.1145/2461121.2461124.

Vigo, M., Harper, S., 2013. Coping tactics employed by visually disabled users on the web. Int. J. Hum.-Comput. Stud. 71 (11), 1013–1025. http://dx.doi.org/10.1016/j.ijhcs.2013.08.002.

Vigo, M., Harper, S., 2014. A snapshot of the first encounters of visually disabled users with the Web. Comput. Hum. Behav. 34, 203–212. http://dx.doi.org/10.1016/j.chb.2014.01.045.

Vollenwyder, B., Buchmüller, E., Trachsel, C., Opwis, K., Brühlmann, F., 2020a. My train talks to me: Participatory design of a mobile app for travellers with visual impairments. In: Miesenberger, K., Manduchi, R., Rodriguez, M.C., Peňáz, P. (Eds.), Computers Helping People with Special Needs, 17th International Conference, ICCHP 2020, Lecco, Italy, September 9–11, 2020, Proceedings, Part I. In: Lecture Notes in Computer Science, Springer Berlin Heidelberg, pp. 10–18. http://dx.doi.org/10.1007/978-3-030-58796-3_2.

Vollenwyder, B., Iten, G.H., Brühlmann, F., Opwis, K., Mekler, E.D., 2019. Salient beliefs influencing the intention to consider Web Accessibility. Comput. Hum. Behav. 92, 352–360. http://dx.doi.org/10.1016/j.chb.2018.11.016.

Vollenwyder, B., Opwis, K., Brühlmann, F., 2020b. How web professionals perceive web accessibility in practice: Active roles, process phases and key disabilities. In: Miesenberger, K., R., M., Rodriguez, M.C., Peňáz, P. (Eds.), Computers Helping People with Special Needs, 17th International Conference, ICCHP 2020, Lecco, Italy, September 9–11, 2020, Proceedings, Part I. In: Lecture Notes in Computer Science, Springer Berlin Heidelberg, pp. 294–302. http://dx.doi.org/10.1007/978-3-030-58796-3_35.

Vollenwyder, B., Schneider, A., Krueger, E., Brühlmann, F., Opwis, K., Mekler, E.D., 2018. How to use plain and easy-to-read language for a positive user experience on websites. In: Miesenberger, K., Kouroupetroglou, G. (Eds.), Computers Helping People with Special Needs, 16th International Conference, ICCHP 2018, Linz, Austria, July 11-13, 2018, Proceedings, Part I. In: Lecture Notes in Computer Science, Springer Berlin Heidelberg, pp. 514–522. http://dx.doi.org/10.1007/978-3-319-94277-3_80.

Waddell, C.D., 2006. Worldwide accessibility laws and policies. In: Web Accessibility: Web Standards and Regulatory Compliance. A Press, pp. 547–579. http://dx.doi.org/10.1007/978-1-4302-0188-5_17.

Web Accessibility Initiative, 2016. Accessibility, usability, and inclusion. URL: https://www.w3.org/WAI/fundamentals/accessibility-usability-inclusion/.

World Health Organization WHO, 2019. World Report on Vision. URL: https://www.who.int/publications/i/item/9789241516570.

World Wide Web Consortium, 1997. World wide web consortium (W3C) launches international WebAccessibility initiative. URL: https://www.w3.org/Press/WAI-Launch.html.

Yesilada, Y., Brajnik, G., Harper, S., 2009. How much does expertise matter?. In: Proceeding of the eleventh international ACM SIGACCESS conference, ACM, http://dx.doi.org/10.1145/1639642.1639678.

Yesilada, Y., Brajnik, G., Vigo, M., Harper, S., 2012. Understanding web accessibility and its drivers. In: W4A '12: Proceedings of the International Cross-Disciplinary Conference on Web Accessibility. p. 19. http://dx.doi.org/10.1145/2207016.2207027.

Yesilada, Y., Brajnik, G., Vigo, M., Harper, S., 2014. Exploring perceptions of web accessibility: a survey approach. Behav. Inf. Technol. 34 (2), 119–134. http://dx.doi.org/10.1080/0144929x.2013.848238.